

HW 4

4.1

Solution :

Here,

$\bar{x}$ = mean

s = deviation

μ = mean of sample population

Given,

$$n = 20$$

$$\bar{x} = 10$$

$$\text{and } s = 1.03$$

$$\alpha = 1 - 99\% = 1 - 0.99 = 0.01$$

for 99% confidence interval

$$\text{and } \alpha_{\frac{1}{2}} = 0.005$$

$$1 \quad df = \text{degree of freedom} = n - 1 = 19$$

Using the table
So, $t_{\alpha_{\frac{1}{2}}, n-1} = t_{0.005, 19} \approx 2.861$

So, the formula becomes

$$\bar{x} \pm t_{\alpha_{\frac{1}{2}}, n-1} \frac{s}{\sqrt{n}}$$

$$10 \pm 2.861 \frac{1.03}{\sqrt{20}}$$

$$= 10 \pm 2.861 \times 0.2303$$

$$= 10 \pm 0.659$$

$$(9.341, 10.659)$$

99% $\pm$ C.I is approximately,
 $(9.34 < \mu < 10.66)$

2.

Here, $G = 5$

$$\text{So, } G = 1.03$$

$$Z_{\alpha/2} = Z_{0.005} \approx 2.576$$

$$\bar{x} \pm Z_{\alpha/2} \frac{G}{\sqrt{n}}$$

$$10 \pm 2.576 \times \frac{1.03}{\sqrt{20}}$$

$$10 \pm 0.5903$$

$$(9.41, 10.593)$$

$$(9.41 < \mu < 10.593)$$